Relativistic Reality

The classical Newtonian worldview was pleasing. Not only did it describe natural phenomena with striking accuracy, but the details of the description—the mathematics—aligned tightly with experience. If you push something, it speeds up. The harder you throw a ball, the more impact it has when it smacks into a wall. If you press against something, you feel it pressing back against you. The more massive something is, the stronger its gravitational pull. These are among the most basic properties of the natural world, and when you learn Newton's framework, you see them represented in his equations, clear as day. Unlike a crystal ball's inscrutable hocus-pocus, the workings of Newton's laws were on display for all with minimal mathematical training to take in fully. Classical physics provided a rigorous grounding for human intuition.

Newton had included the force of gravity in his equations, but it was not until the 1860s that the Scottish scientist James Clerk Maxwell extended the framework of classical physics to take account of electrical and magnetic forces. Maxwell needed additional equations to do so and the mathematics he employed required a higher level of training to grasp fully. But his new equations were every bit as successful at explaining electrical and magnetic phenomena as Newton's were at describing motion. By the late 1800s, it was evident that the universe's secrets were proving no match for the power of human intellectual might.

Indeed, with the successful incorporation of electricity and magnetism, there was a growing sense that theoretical physics would soon be complete. Physics, some suggested, was rapidly becoming a finished subject and its laws would shortly be chiseled in stone. In 1894, the renowned experimental physicist Albert Michelson remarked that "most of the grand underlying principles have been firmly established" and he quoted an "eminent scientist"—most believe it was the British physicist Lord Kelvin—as saying that all that remained were details of determining some numbers to a greater number of decimal places.[1] In 1900, Kelvin himself did note that "two clouds" were hovering on the horizon, one to do with properties of light's motion and the other with aspects of the radiation objects emit when heated,[2] but there was a